LAB EXPERIMENT RECORD

SET 16 – Customer Purchase

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 16 – Customer Demographics Analysis

To visualize customer demographic details, including age, gender, and income, using R programs and an interactive dashboard in Tableau.

Dataset: Customer Demographics

Customer ID	Age	Gender	Income ($)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

PROGRAM 1: Bar Chart (Customer Age Distribution)

Aim: To display the age distribution of customers using a bar chart**.**

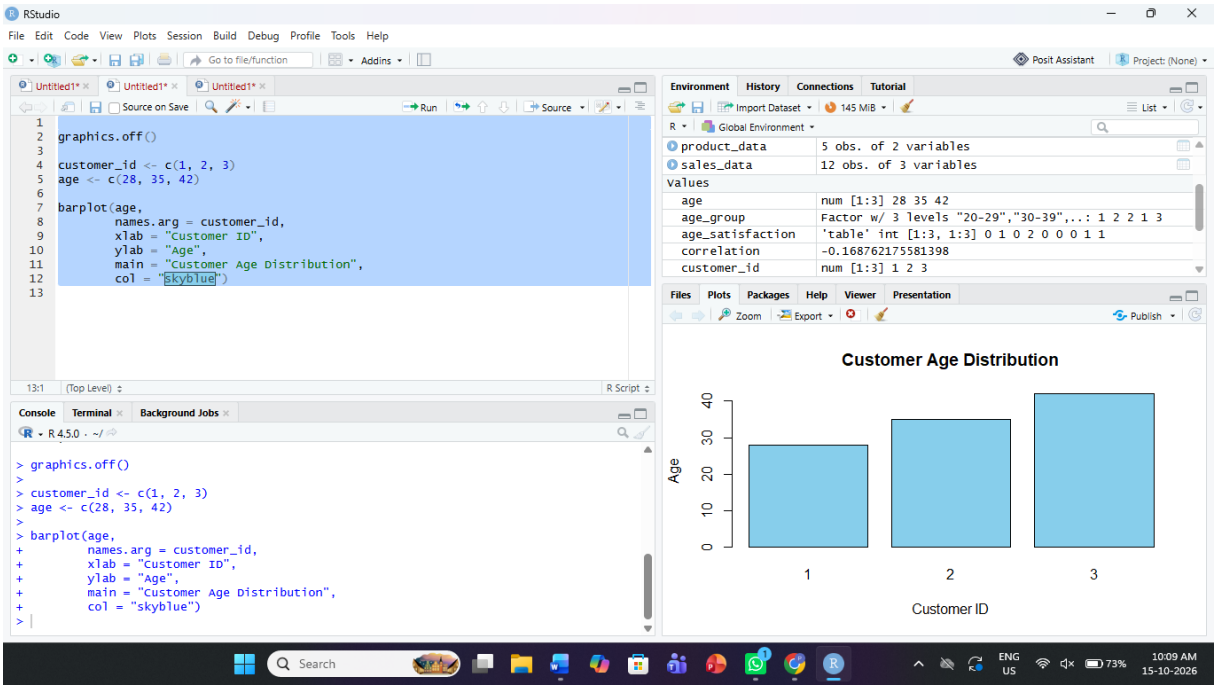
R Program:

```
graphics.off()

customer_id <- c(1, 2, 3)

age <- c(28, 35, 42)
```

```
barplot(age,
        names.arg = customer_id,
        xlab = "Customer ID",
        ylab = "Age",
        main = "Customer Age Distribution",
        col = "skyblue")
```



Observation: The bar chart shows the ages of three customers. Customer 3 has the highest age of 42 years, while Customer 1 has the lowest age of 28 years**.**

PROGRAM 2: Pie Chart (Gender Distribution)

Aim: To represent the gender distribution of customers using a pie chart**.**

R Program:

```
graphics.off()
```

```
gender_count <- c(2, 1)

gender <- c("Female", "Male")

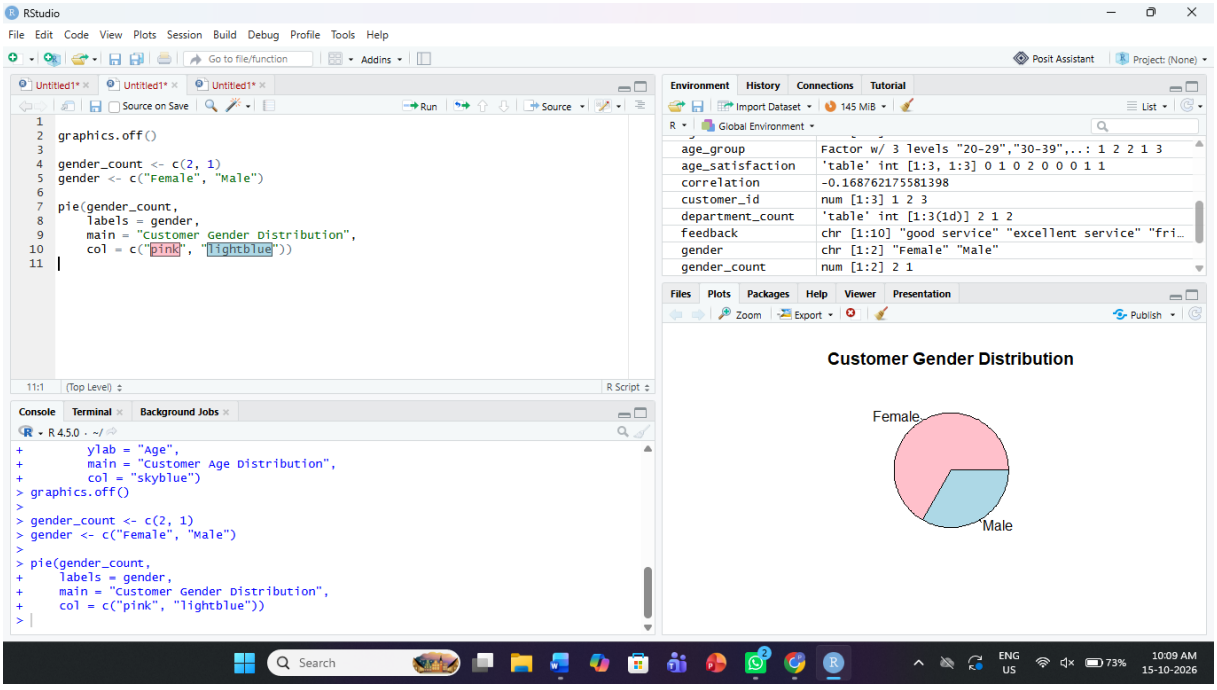


pie(gender_count,

   labels = gender,

   main = "Customer Gender Distribution",

   col = c("pink", "lightblue"))
```



Observation: The pie chart shows that two customers are female and one customer is male.

Females represent approximately 66.7% of the customers

PROGRAM 3: Customer Demographics Summary Table

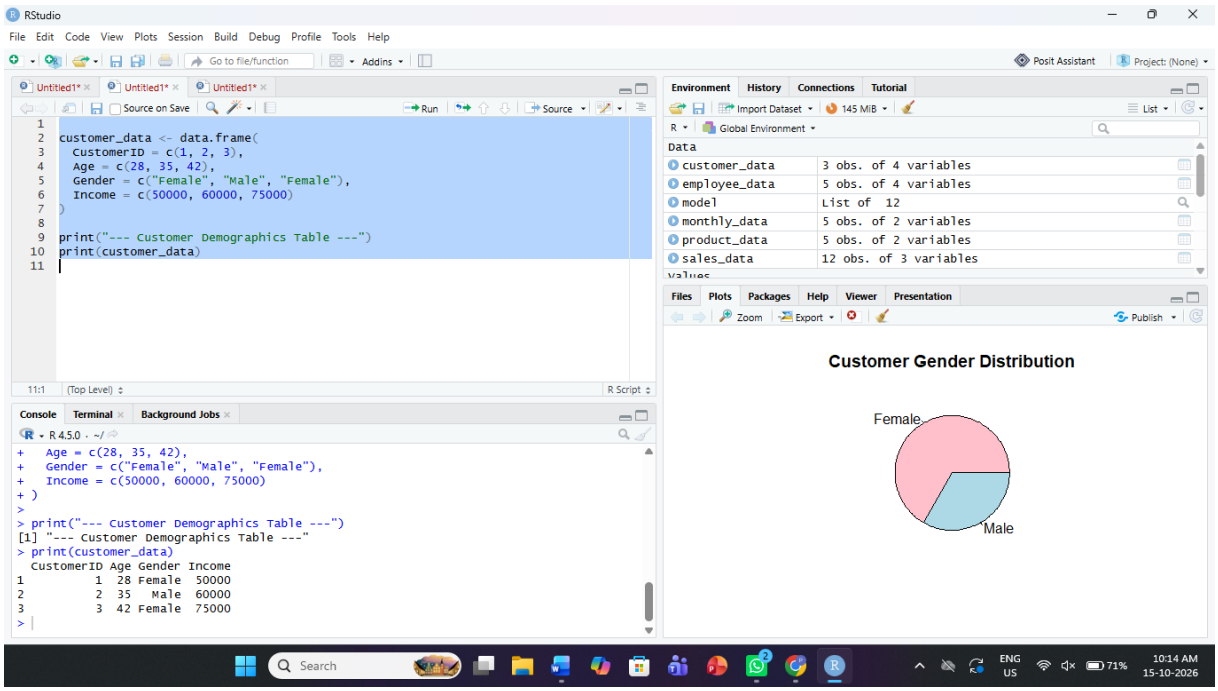
Aim: To create a structured table summarizing customer demographic information**.**

R Program:

```
customer_data <- data.frame(
  CustomerID = c(1, 2, 3),
```

```
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

print("--- Customer Demographics Table ---")
print(customer_data)
```



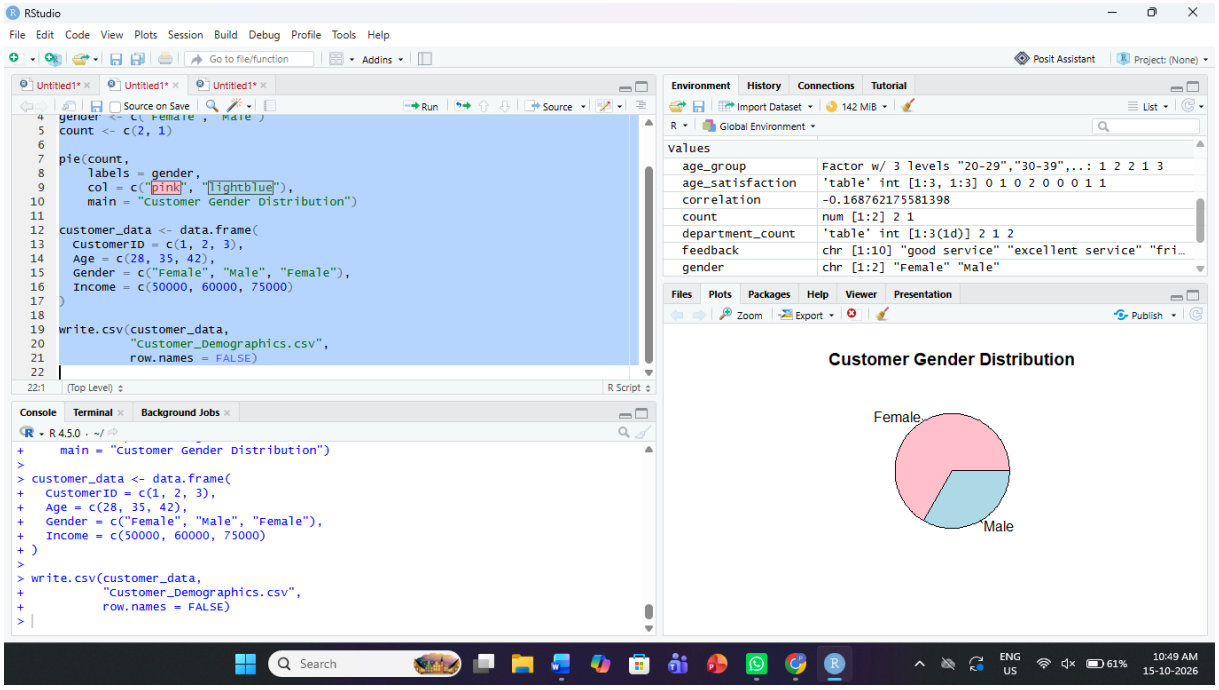
Observation: The table displays customer ID, age, gender, and income. Customer 3 has the highest income of $75,000.

PROGRAM 4: Customer Demographics Dashboard in Tableau

Aim: To create an interactive Tableau dashboard combining customer age, gender distribution, and demographic records**.**

Steps in Tableau:

1. Save the dataset as Customer_Demographics.xlsx and connect it to Tableau.
2. Sheet 1 – Age Distribution: Drag Customer ID to Columns and Age to Rows. Select Bar Chart.
3. Sheet 2 – Gender Distribution: Drag Gender to the view and use the gender count to create a pie chart.
4. Sheet 3 – Customer Details: Display Customer ID, Age, Gender, and Income in a table.
5. Create a new dashboard and arrange all three sheets.
6. Enable Use as Filter to interactively filter customer information**.**



Observation: The dashboard displays customer age, gender, and income in one place. Filtering helps users focus on selected customer records.

Procedure:

1. Create vectors containing customer demographic details.
2. Use barplot() to display customer ages.
3. Use pie() to visualize gender distribution.
4. Create a data frame to display customer information.
5. Build and arrange the visualizations in a Tableau dashboard**.**

Result:

Customer demographic data was visualized using bar charts, a pie chart, and a summary table in R. An interactive dashboard was designed in Tableau**.**